AIML-A05

Privacy-Preserving Federated Learning Framework for Early Brain Tumor Classification Using Multi-Hospital MRI Images

Batch ID	AIML-A05	**Category**	Not ticked on form (department rule: Research oriented)
Team	24BQ5A6120: A. Venkata Vaibhav 23BQ1A6127: D. Vinay Kumar 23BQ1A6141: G. Lasya Deekshitha 23BQ1A6117: B. Aasritha Raj	**Guide**	Dr. Mahammad Firose Shaik
Base paper	Albalawi et al., Integrated Approach of Federated Learning with Transfer Learning for Classification and Diagnosis of Brain Tumor. BMC Medical Imaging 24:110, 2024	**Supporting**	Lee et al., Federated learning AI platform for pediatric brain tumors (Nature Communications, 2024) Lhasnaoui et al., Decentralized Diagnosis (KDD FL Workshop, 2024) Ay & Ekinci, CNN-based privacy-preserving FL (IJIST, 2024) Gohari et al., FedBrain-Distill (arXiv:2409.05359, 2024)

Hierarchical Implementation Modules

ID	Module	Sub-modules	Key tools / techniques	Math to present (Review II)	Deliverable / checkpoint	Timeline
M0	**Foundations: Medical Image DL & Federated Learning**	0.1 CNNs and transfer learning (ResNet50, EfficientNetB0) 0.2 Brain MRI basics: T1, T1-contrast, T2, FLAIR; appearance of glioma, meningioma, pituitary tumours 0.3 Federated learning concepts: clients, server, rounds, FedAvg 0.4 GPU environment and experiment tracking	PyTorch, torchvision / timm, Colab / Kaggle GPU, Weights & Biases	Convolution; softmax cross-entropy; FedAvg intuition	Each member fine-tunes a pretrained CNN on a small MRI subset + completes the FL short course	Week 1

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M1	**Datasets & Multi-Hospital Simulation Design**	1.1 Download Kaggle Brain Tumor MRI (7,023 images) and BraTS 2021 1.2 Duplicate / near-duplicate detection (perceptual hash); patient-level grouping 1.3 Extract 2-D tumour slices from BraTS NIfTI volumes (T1ce / FLAIR) 1.4 Hospital scenarios: (a) IID random, (b) natural split by source, (c) Dirichlet label skew α = 0.1 / 0.5 / 1.0, (d) BraTS glioma-only hospital 1.5 Fixed global test set + per-hospital local test sets	pandas, imagehash, NiBabel, Flower Datasets partitioners	Dirichlet partitioning Dir(α); class-distribution entropy per client	Data card + partition files for every scenario	Weeks 1–2
M2	**Preprocessing & Harmonisation**	2.1 Crop to brain region, resize to 224×224, convert grayscale → 3 channels 2.2 Intensity normalisation per image; histogram matching (optional) to reduce scanner differences 2.3 Augmentation: flips, small rotations, brightness / contrast 2.4 Per-client data loaders; class-weighting for imbalance	OpenCV, Albumentations, MONAI transforms	Z-score normalisation; histogram matching; class weights $w_k = N / (K \cdot n_k)$	Reusable preprocessing + loader module used by all clients	Weeks 2–3
M3	**Centralised & Local-only Baselines**	3.1 Centralised training (all data pooled) = upper bound 3.2 ResNet50 vs EfficientNetB0 (+ VGG16 to reproduce the base paper) 3.3 Feature extraction vs fine-tuning 3.4 Local-only training per hospital = lower bound (shows why collaboration helps) 3.5 Fix hyper-parameters for fair FL comparison	torchvision / timm pretrained models, PyTorch Lightning (optional)	Transfer learning; precision / recall / F1; model size (params) vs accuracy	Baseline table: centralised, local-only, base-paper reproduction	Weeks 3–5
M4	**Federated Training with FedAvg**	4.1 Flower simulation: 4–10 hospital clients 4.2 Client training loop (local epochs E, batch size B); server FedAvg 4.3 Partial participation (fraction of clients per round), number of rounds 4.4 IID vs natural-source vs Dirichlet non-IID experiments 4.5 Track global and per-client accuracy per round	Flower (flwr), PyTorch	FedAvg $w = \Sigma_k (n_k/n) \cdot w_k$; effect of E and client fraction C on convergence	FedAvg matching centralised accuracy within a stated margin on IID data	Weeks 4–6

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M5	**Handling Heterogeneous (Non-IID) Hospitals**	5.1 FedProx (proximal term for client drift) 5.2 FedBN (keep BatchNorm layers local; suits scanner differences) 5.3 Server optimisers: FedAdam / FedYogi 5.4 Optional: SCAFFOLD or personalised fine-tuning per hospital 5.5 Compare all on the non-IID scenarios from M1	Flower strategies, Flower Baselines (FedBN, FedProx)	FedProx loss $L + (\mu/2)\|\|w - w^t\|\|^2$; BatchNorm statistics; client drift	Table: FedAvg vs FedProx vs FedBN vs FedAdam under non-IID	Weeks 6–8
M6	**Privacy & Security Layer**	6.1 Differential privacy on client updates (clipping + Gaussian noise); ε sweep 6.2 Secure aggregation (SecAgg+) so the server never sees individual updates 6.3 Empirical privacy test: membership-inference attack with / without DP 6.4 Basic robustness: detect a faulty or malicious hospital (median / trimmed mean)	Opacus or Flower DP wrappers, Flower SecAgg+ mod, ML Privacy Meter	(ε, δ)-DP; Gaussian mechanism; clipping norm C; attack AUC	Privacy–utility curve (accuracy vs ε) + attack results	Weeks 7–9
M7	**Communication Efficiency**	7.1 Measure bytes per round and total to reach target accuracy 7.2 ResNet50 (~25M params) vs EfficientNetB0 (~5M) cost comparison 7.3 More local epochs / fewer rounds trade-off 7.4 Update compression: 8-bit quantisation or top-k sparsification 7.5 Optional: knowledge-distillation approach (FedBrain-Distill idea)	Flower metrics, PyTorch quantisation utilities	Communication cost = rounds × clients × model size; compression ratio	Accuracy-vs-communication plot for each model and method	Weeks 8–9
M8	**Evaluation & Explainability**	8.1 Accuracy, precision, recall, F1 (per class + macro), confusion matrix, ROC-AUC 8.2 Federated vs centralised vs local-only across all scenarios 8.3 Per-hospital fairness: worst-hospital accuracy 8.4 External test on BraTS glioma slices (generalisation) 8.5 Grad-CAM heatmaps; check focus on tumour region 8.6 Repeated runs (3 seeds), mean ± SD, statistical tests	scikit-learn, torchmetrics, pytorch-grad-cam, SciPy	Macro-F1; ROC-AUC; Grad-CAM; paired t-test / Wilcoxon	Final results tables, convergence plots, Grad-CAM gallery	Weeks 9–11

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M9	**Hospital Dashboard, Demo & Reporting**	9.1 Dashboard: rounds, global / per-hospital accuracy, ε spent, bytes sent 9.2 Diagnosis demo: upload MRI → class, confidence, Grad-CAM 9.3 Multi-container demo (server + hospital clients) with Docker Compose 9.4 Final report, paper draft, reproducible code	Streamlit, Docker Compose, Flower deployment	—	Live demo + report	Weeks 10–12